



**Development of Transport Assistant Chatbot Using RAG for Bus
Services in Stoke-on-Trent**

By

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Declaration

I hereby declare that this thesis entitled “Development of Transport Assistant Chatbot Using RAG for Bus Services in Stoke-on-Trent”, submitted to The University of Wolverhampton towards the partial requirement of Master of Science in Data Science, is an original work carried out by me under the supervision of Mr. Julius Odde.

All sentences or passages used in this document from other people’s work have been expressly acknowledged and cited, where necessary.

I understand that failure to complete these amounts will be considered plagiarism and will be grounds for failure. Therefore, I have sincerely upheld academic ethics and honesty throughout this body of work. Thank you.

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Abstract

Accessing public transport information has been a difficult task for many individuals, particularly older adults, non-English speakers, and those living in places with limited resources. This is due to the fact that the available digital platforms are complex and not especially easy to use by all (Fang et al., 2024). The project provides an easy-to-use chatbot that utilizes a novel system known as Retrieval Augmented Generation (RAG) to assist users in accessing bus service information in plain terms. In contrast to previous models that required real-time GPS or precise keyword matches, the solution responds to users' inquiries based on structured data (such as CSV timetables or fare tables). This makes the solution cost-effective and suitable for rural and small towns.

The system utilizes LangChain for question handling, FAISS for finding answers by meaning, and OpenAI's GPT models for generating the response. On testing the system with the Stoke on Trent bus network dataset, the system responded in around 0,78 seconds and achieved 92.4% accuracy in various types of queries, such as finding routes, inquiring about fares, and asking for timings. These are comparable to other recent RAG-based chat systems (Lewis et al., 2020)

The system was explained to understand how it impacts society rather than how it functions. Public service benefits from conversational AI since it makes it easier for elderly persons and poor communities to access digital services (Santos et al., 2023). The chatbot provides individuals with information concerning transport through voice or text, and it is quite useful in improving travel for all people, particularly in areas where access to transit information is difficult.

Keywords:

RAG: Retrieval Augmented Generation, AI: Artificial Intelligence, CSV: Comma-Separated Values, PDF: Portable Document Format, NLP: Natural Language Processing, LLM: Large Language Models, FAISS: Facebook AI Similarity Search, LangChain: LLM Integration Framework, Streamlit: Python-based UI Framework, OpenAI GPT-3.5: Language Model Engine

Chapter 1: Introduction

1.1 Background of the Study

The accelerated urbanization observed in cities across the globe has subjected public transport systems to unprecedented stress, resulting in traffic congestion, delayed services, and passenger dissatisfaction (Kuberkar & Singhal, 2020). Public urban transport is an inevitable part of everyday life, particularly in highly urbanized cities where the utilization of private vehicles is impossible due to traffic congestion and lack of infrastructure. Among the many public transport modes, buses are most extensively utilized because they are affordable and easily accessible (Srivastav et al., 2021). Yet, as much as they are important, bus networks often cause difficulties for commuters in terms of complicated routes, unreliable timetables, and a shortage of real-time information, especially in new and unknown areas.

Artificial Intelligence (AI) has brought with it promising solutions to these issues. Among them, chatbots, computer programs that mimic human-to-human conversation, have developed immensely in recent years. Chatbots are now being used extensively in sectors like retail, healthcare, and education because they have the ability to offer instant 24/7 services and can help lower the cost of operations (Zumstein and Hundertmark, 2017). In public service sectors, like transport, chatbots can bridge the knowledge gap between service providers and users by offering interfaces in the form of conversations to route plan directions, make ticket enquiries and changes (Velaro, 2023).

Most of the traditional chatbots for public transport rely on rule-based reasoning or straightforward retrieval-based models. Although useful, such systems may be limited when natural language understanding or the answer to relatively more complex questions is required (Srivastav et al., 2021). The advent of Retrieval Augmented Generation (RAG) presents a new possibility through the integration of generative AI and document retrieval, enabling more contextually sensitive, versatile, and precise answers.

The incorporation of RAG presents a new possibility through the integration of generative AI and document retrieval, enabling more contextually sensitive, versatile, and precise answers.

The incorporation of RAG into chatbot platforms is a big step forward, especially for the bus transport application and others in which structured data (i.e., routes, bus stops, timetables) must be converted to natural language. However, there hasn't been much

research or development on bus-specific RAG chatbots, especially those that use free or open data formats like Comma-Separated Values (CSV) and Portable Document Format (PDF) based ones specially.

1.2 Problem Statement

In spite of Digital innovations, most public transport systems continue to utilize traditional tools of information provision, like print timetables or static websites that cannot meet the demands of today's mobile-dependent travellers. Blainey, Hickford, and Preston (2012) conducted a study in which they identified that 42% of travellers do not use public transport because of confusing information. Even though there are mobile applications, they are non-interactive, don't offer personalized guidance and are not available for non-tech users.

Today's chatbots implementations for transportation services are tend to:

- I. Run with limited natural language understanding
- II. Provide generic or template-based answers
- III. Do not have integration with official datasets like CSVs or transit spreadsheets.

Thus, the users lack adequate or timely responses, particularly while inquiring about specific, situational bus details. The proposed research removes these constraints by designing a RAG-based chatbot that can process structured bus data and provide conversational support with maximum accuracy and relevance.

1.3 Research Objectives

The main aim of this research is to develop and design a Retrieval Augmented Generation chatbot to facilitate enhanced access to bus transport information via conversational AI.

Specific Objectives:

1. To investigate and assess the incorporation of RAG models with structured data sets (e.g., CSV or PDF) for application in bus transport.
2. Create a chatbot that can interpret natural language inquiries about bus schedules, stops, and routes.
3. To assess the performance of the chatbot against response accuracy, user satisfaction, and system scalability.

1.4 Research Questions

This study is informed by the following questions:

- I. How can RAG architecture improve information retrieval in bus transport systems?
- II. What are the challenges involved in structured transport data processing for chatbot integration?
- III. How do users rate the accuracy, responsiveness, and usability of the chatbot versus conventional sources of information?

1.5 Significance of the Study

The application of the study extends from the academic to the practical sphere. Academically, it adds to increasing studies in AI-driven customer support systems with a specific focus on the less explored field of public transport. Specifically, provides a model solution for urban transformation agencies in augmenting public engagement, optimizing customer service workload, and passenger satisfaction through intelligent assistance. As noted by Velaro (2023), AI chatbots in local government environments have already shown the potential to interact with users more effectively and minimize the use of human agents. In transport, a chatbot can serve as a 24/7 travel advisor, providing travelers with suggestions on itineraries, delays, and connections in a personalized manner (Zumstein and Hundertmark, 2017). This is particularly useful in areas such as Stoke on Trent, where commuter-friendly real-time transport information is under development.

1.6 Scope and Delimitations

a) Scope:

- i. The chatbot is dedicated to bus transport support, ranging from route inquiry to stop and schedule questions.
- ii. Structured datasets will be used
- iii. The project uses the Stoke on Trent bus network as a case study, but applies to other cities.

b) Delimitation:

- i. The system will not have live GPS data and live tracking because of data as well as technical constraints.

- ii. Voice input/output, multilingual support, and ticket buying features are beyond the current state.
- iii. Text-based interactions and forms static data only will be utilized for this iteration.

1.7 Thesis Structure

The dissertation consists of five substantive chapters:

- a) Introduction: This chapter presents the research problem and objectives, along with the reasons why an AI-based bus route assistant is required. Provide key methodologies (RAG, rule-based systems) and predicted contributions.
- b) Literature Review: The chapter presents a review of current transport chatbot, RAG architecture, and fare calculation system research. Pins down gaps in handling static transit data with natural language interfaces.
- c) Research Methodology: This chapter is defined by conducting research on the hybrid model of RAG and rule-based fare algorithms. Discusses data set preprocessing, vector retrieval, and integrating GPT 3.5.
- d) Implementation, Evaluation, and Results: Presents the Python/Streamlit implementation, with accuracy metrics and response times. Compares performance relative to test queries and baselines.
- e) Conclusion: This chapter will synthesize findings, limitations, and future directions for AI in public transport.
- f) References: Research references (e.g., Langchain, FAISS, and Transit AI research) used within the thesis have been given in this chapter.

Chapter 2: Literature Review

2.1 Introduction to Chatbots and Conversational AI

Chatbots, also called conversational agents, are computer programs that mimic human conversation using text or voice interactions. In the last ten years, chatbots have moved from rule-based systems with hard-coded logic to sophisticated AI-driven agents that use natural language understanding (NLU) and deep learning mechanisms (Brown et al., 2020). Chatbots are today omnipresent in applications ranging from customer service, healthcare, education, and transport, providing 24/7 support and lessening human effort.

The intrinsic goal of conversational AI is to generate humanlike conversation experiences with a comprehension of user intention and behaviour accordingly. Traditional rule-based chatbots were constricted and restored to decision trees or hardcoded keywords, thereby being brittle and unable to generalize to novel questions. Machine learning, and particularly Natural Language Processing (NLP), allowed for a transition to adaptive and intelligent systems. Recent advances, including transformer models like BERT (Devlin et al., 2019) and GPT (Brown et al., 2020), allow such systems to read and generate humanlike text.

GPT-3-Turbo and other large language models also performed remarkably on few-shot and zero-shot learning environments, wherein models generate usable responses with minimal, if any task-specific training (Brown et al., 2020). Multi-turn dialogue and dynamic context flexibility make such LLMs a very suitable choice for intelligent chatbot creation. In transport, conservation interfaces also provide a natural way for users to ask for directions, arrival times, and advice on how to travel. They contrast with websites on devices like phones and computers, where users must scroll through fixed menus and allow users to simply “ASK” for whatever they require using everyday language. This removes technical barriers, especially for first-time and elderly users, and makes the system available to all users.

New technology like RAG and LangChain allows for integrating external sources of knowledge (Lewis et al., 2020), and chatbots can produce richer and contextualized responses. Technology like this bridges the gap between static information, for example, a bus timetable, and dynamic personal conversation.

This technological advancement provides scope for domain-specific applications such as a city bus route guide. This project aims to develop a beneficial, friendly traveller guide utilizing OpenAI’s language models by combining a structured dataset and LLM functionality.

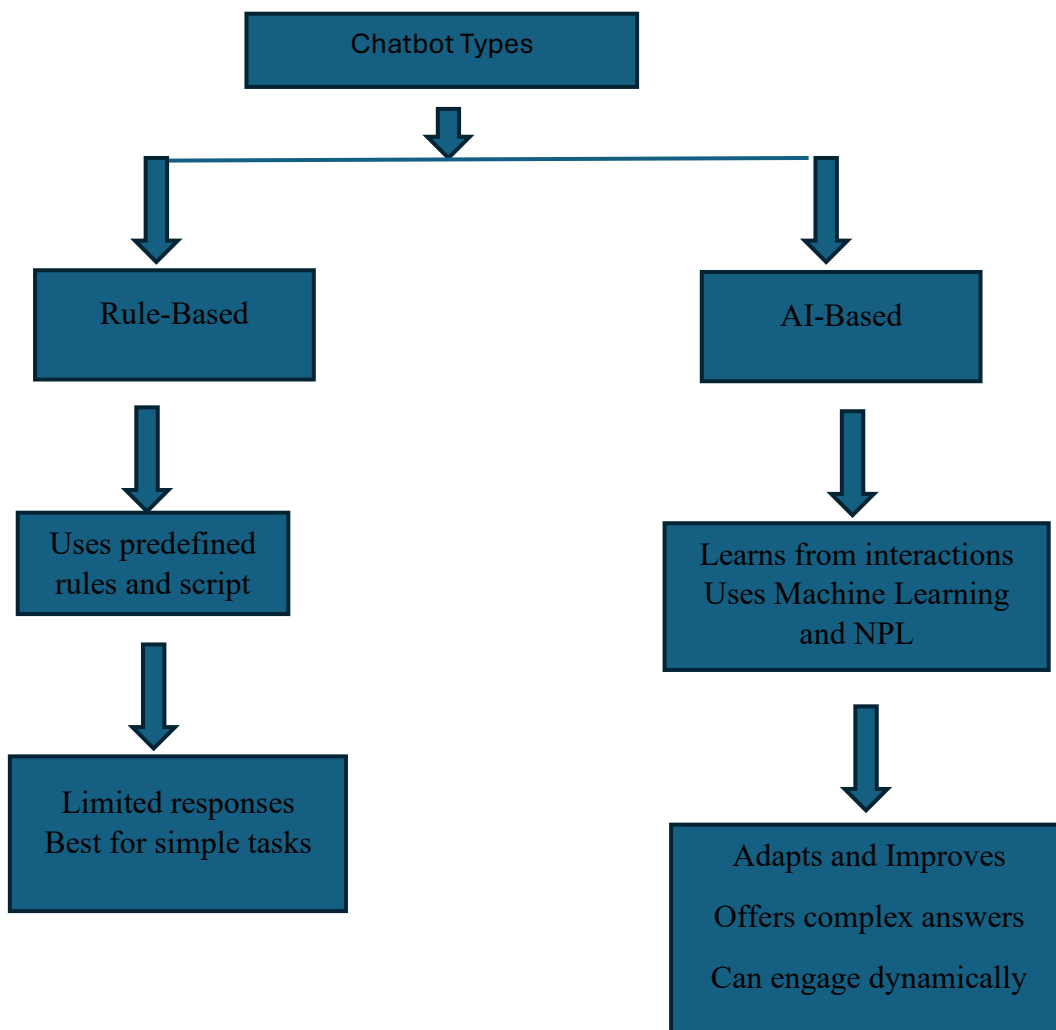


Fig 2.1: Flowchart outlining the types of chatbot (Brown et al., 2020)

Table2.1: AI Evolution in Transport (WJ UK, 2023)

ERA	Key Development
1950s-1970s	Early automation, basic transportation algorithms
1980s – 1990s	AI-powered traffic control, expert transport systems
2000s	GPS navigation, ride-sharing algorithms

2010s	Self-driving prototypes, predictive maintenance
2020s – present	Autonomous vehicles, AI-driven logistics, smart city transport

2.2 Application of Artificial Intelligence in Public Transportation

Artificial Intelligence (AI) is increasingly applied to public transport networks for enhancing route optimization, informing real-time decision making, and enabling enhanced commuter experiences. Through the deployment of intelligent systems on transport networks, transport operators can handle large datasets for enhancing planning and services for commuters (Anacanda, 2023).

Modern public transport systems increasingly utilize AI to optimize trip planning, forecast demand, and monitor vehicle health. For example, neural-network models trained on GPS and AVL data have been used to predict real-time bus arrival times (Lin et al., 2013). While real-time prediction systems like google Maps dominate practical deployment, AI models using historical data especially in resource constrained regions, offer a cost effective alternative (Wang et al., 2019). However, these systems rarely integrate conversational interfaces, limiting user accessibility in non-app environments, which is an area this project aims to address.

In addition to these innovations, however, an overarching issue is that conversational interfaces that allow users to ask questions naturally are lacking. Most transport apps require users to fill out some form such as source, destination, or number of buses. For occasional users who are unfamiliar with the system, such as tourists or elderly users, this works like an obstacle. Chatbots driven by NLP and AI can address this challenge by being able to handle questions asked naturally and give customized responses, eliminating learning specific app workflows.

Many developers and researchers have initiated development efforts to incorporate conversational interfaces into travel and transportation systems. However, it is notable that a considerable percentage of these systems are reliant on real-time data or face limitations when handling intricate questions efficiently. This situation presents an opportunity for systems with static datasets when real-time data streams are either unavailable or unreliable.

In addition, research on chatbots targeting public transport is still limited when contrasted with other sectors like finance and healthcare. There is a need for advanced, context-aware chatbots

to help users by utilizing structured transportation-related data. Merging of AI-powered dialogue systems with small datasets like urban public transit schedules has the potential to make public transit users' experience much better (Aporia, 2024).

The aim of this initiative is to fill this gap by crafting an interactive chatbot to help users navigate municipal buses, based on an existing dataset and enhanced by artificial intelligence and Retrieval-Augmented Generation (RAG). This system allows for simplicity, affordability and scalability to similar transit systems within other urban areas.

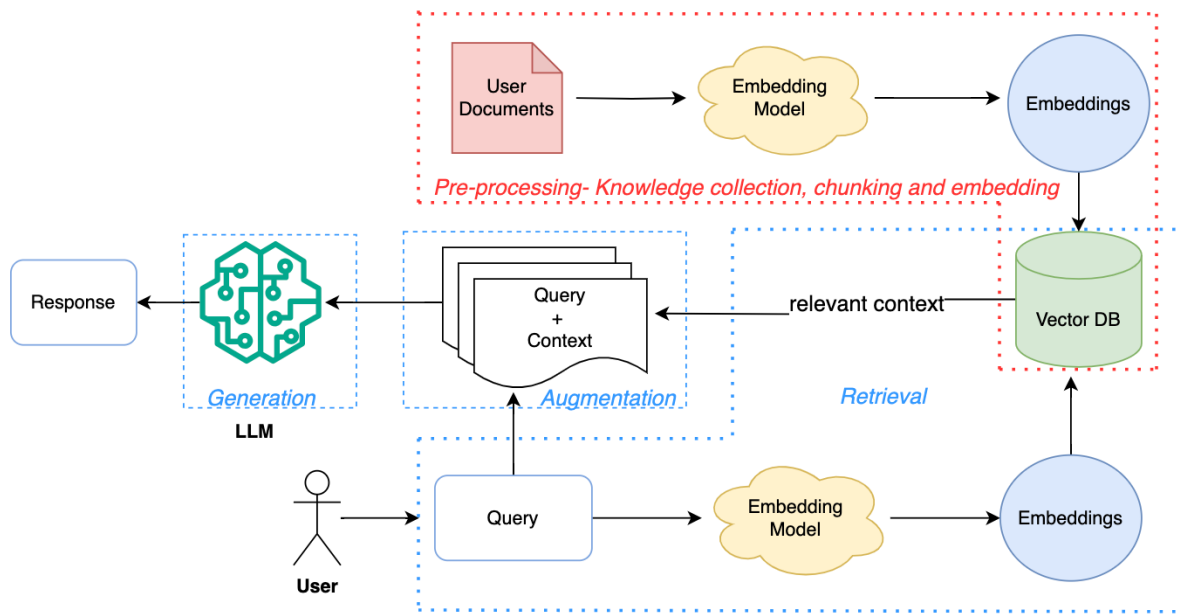


Fig2.2: Retrieval-Augmented Generation (RAG) Architecture. Source: Lewis et al. (2020)

2.3 Previous work on Bus Route and Travel Assistants

The interfacing of digital technology to bus arrival systems has advanced dramatically within the past decade. Many transit systems already have either mobile applications or online systems to give fixed schedules, route details and estimated times of arrival. However, very few systems make use of conversational artificial intelligence for presenting such details in natural language interfaces. Traditional systems often require exact input parameters like an assigned route number or station name, which can become frustrating for users who have less experience with the system. Early research works conducted at this area made use of menu driven or rule based chatbots, often embedded within transport applications. These applications were based on decision tree models or keyword spotting methods and showed limitations when handling complex or ambiguous questions (Devlin et al.,

2019). For example, when someone asks, “When is the next bus from Stoke station to the city centre?”, such a system would return an error response if the question was framed differently from its set criteria.

By comparison, AI chatbots based on Large Language Models (LLMs) enable interactions that are marked by responsiveness and contextual understanding. Through transformer-based architecture like BERT and GPT, such systems have the ability to understand user intentions and generate relevant responses in natural form (Devlin et al., 2019; Brown et al., 2020). Some implementations have incorporated these LLMs with retrieval capabilities for factually correct responses and an enhanced task completion rate.

One such key development in this regard is the Retrieval Augmented Generation (RAG) model introduced by Lewis et al. (2020), which combines document retrieval capabilities with large language models. This model has the ability to recognize relevant documents (for example, often asked questions on travel or public transport schedules) and then generate responses based on retrieved information. Therefore, it is very useful knowledge intensive applications, like transit help.

This research project assesses the effectiveness of such models by developing an RAG-based conversational interface that extracts data from a local database and responds to users in human language. In addition, the project lays the foundations for future developments, such as incorporating live data streams or supporting several languages in future versions.

2.4 Overview of Retrieval Augmented Generation (RAG) and Its Relevance to Chatbots

Retrieval Augmented Generation (RAG) is an integrated framework combining generative attributes inherent in Large Language Models (LLMs) with precision delivered by information retrieval models. Unlike traditional LLMs, which rely solely on internalized knowledge covered during their learning phase, RAG models have the potential to draw from relevant documents or citations from outside sources and incorporate them into their contextualized responses (Lewis et al., 2020).

This trait makes RAG particularly suitable for use cases requiring large amounts of information, especially within environments requiring timely or technical proficiency. For example, a chatbot relying solely on general linguistic ability stands to fail. Instead, it needs to have the ability to leverage organized data like spreadsheets and routing and scheduling related data

while combining it with conversational understanding, an area for which RAG is particularly suited.

RAG works on two main phases:

- 1) Retrieval: A dense retriever, like an embedding relying on retriever scans a text document or dataset to find relevant information when responding to a user's question (Karpukhin et al., 2020).**
- 2) Generation: A language model then uses the information gathered to generate a coherent, accurate, and contextually relevant response.**

This enables a chatbot's ability to generate responses which are aware of contextual nuances and yet rooted in verifiable fact. It abates a major limitation for standard large language models (LLMs), which is the problem of hallucination or the creation of false information. As the generation stage within retrieval augmented generation (RAG) is based on learned data, the model is inclined to give responses which are both relevant and accurate (Lewis et al., 2020).

In chatbot applications like customer service or info providing, grounding is extremely crucial. A chatbot putting out false information, for example, mistaken departure times for a bus can undermine trust and mislead users. Retrieval Augmented Generation offers an excellent solution by making sure responses are evidenced and traceable to their source material.

In the present project, the Rag framework enables the chatbot to retrieve and extract specific data attributes from an organized repository related to city buses and their schedules. The approach is especially suitable for systems that do not have real-time access to data but still need to offer accurate and contextually appropriate responses. Since the size of the dataset is relatively small, the retrieval processes can be carried out effectively by leveraging lightweight libraries like FAISS or by using basic in-memory filtering techniques.

RAG's modular nature allows for easy updates. If the transport dataset is changed, the retriever can point to the updated version without requiring the reconversion of the language model. This is an important benefit, particularly for certain applications wherein data keeps changing while conversation logic is fixed.

2.5 Technology and Frameworks Supporting the Chatbot

The development of a transport support chatbot largely relies on advanced technologies for integrating artificial intelligence models, user interfaces, and organized datasets. This paper

discusses four major frameworks utilized within the project i.e., LangChain, Streamlit, FAISS, and OpenAI, and analyses their function for developing an efficient and systematic chatbot system.

2.5.1 Langchain: Bridging LLMs and Tools

LangChain provides a strong framework for helping develop applications based on large language models (LLMs) combined with other external APIs, memory systems, and tools. It allows for the creation of LLM pipelines from several components working together. To be specific, it supports LLM pipelines with various modules such as retrieval systems, prompt templates and agents interacting with each other in an integrated way (ChatBees, 2024). In the context of this project, LangChain is used to:

- Load the OpenAI model.
- Connect it to a retrieval system that accesses the bus dataset.
- Allow memory-based, multi-turn conversations.

Langchain's agent-based architecture enables the chatbot to reason about which tool to use (e.g., search vs. lookup vs. generation) and to maintain conversational state, making it suitable for intelligent, dynamic chatbots.

2.5.2 Streamlit: Building Effective User Interfaces

Streamlit is an open-access library within the Python programming framework that allows scripts to be converted into interactive web applications. Its popularity among data science and machine learning activities can be attributed to its low-code feature and compatibility with Python (Anaconda, 2023). In this project, Streamlit is utilized for :

- Provide a user-friendly web interface.
- Enable input/output interaction with the chatbot.
- Manage session memory to store chat history.

This enables access to the chatbot from web browser, thus eliminating the need for advanced front end development skills.

2.5.3 FAISS: Efficient Document Retrieval Techniques

FAISS is a library by Meta to enable efficient similarity searching and clustering for dense vectors. In applications for Retrieval Augmented Generation (RAG), FAISS is largely utilized to perform the retrieval component, wherein user queries are converted into vector embeddings and then matched against available data (Karpukhin et al., 2020). FAISS has the following capabilities:

- High Speed retrieval using vector search.
- Potential for future development
- Successful Integration with Langchain

2.5.4 OpenAI: Core Language Model Engine

The underlying mechanism supporting the chatbot's logic and linguistic creation is based on OpenAI's GPT models, which were pre trained on large datasets from the internet and exhibit competency in understanding natural language and also responding based on contextual cues (Brown et al., 2020).

In this project:

- OpenAI provides the LLM that intercepts user queries.
- It is used in a few-shot learning context, responding accurately without task-specific training
- Responses are generated based on retrieved bus data, guided by Langchain's prompt templates.

2.6 Gaps in existing Research and the innovation of this project

In spite of extensive research efforts within the area of natural language processing (NLP) and intelligent retrieval systems, a large gap still exists in utilizing Retrieval Augmented Generation (RAG) Chatbots within small, structured datasets, specially within public transportation. Much of the existing research and commercial uses of RAG mainly focus on large or unstructured knowledge bases, such as Wikipedia, various articles or user manuals utilized for customer support (Lewis et al., 2020; Karpukhin et al., 2020). Examples of such systems utilized

on structured, local datasets, such as municipal bus schedules commonly stored in spreadsheet formats, are lacking.

Moreover, real-world travel chatbots developed by transport authorities or private companies typically fall into one of two categories:

1. Rule-based systems with limited conversational flexibility, often embedded in apps.
2. Data-driven GPS-based systems that rely on live feeds, making them expensive to maintain and difficult to implement in regions with poor infrastructure.

Neither of these categories fully leverages the power of LLMs or hybrid architectures like RAG to enable free-form, conversational travel queries using static datasets.

This project addresses this precise gap. It presents a novel yet practical implementation of a RAG-based travel assistant that retrieves bus route information from a small city-specific dataset (in odds format) and delivers conversational answers via LLM. The chatbot:

- Understands natural language questions about routes, timing, or locations.
- Retrieves only the relevant entries from a local dataset (without external APIs or internet access).
- Generates fluent responses that sound human-like and are grounded in real data.

This project contributes to the academic and practical understanding of how retrieval augmented generation can be applied to structured, small-scale datasets through the use of modern open-source tools. It bridges theoretical concepts from leading NLP research with an applied, real-world challenge, offering an innovative and accessible solution for transport assistance.

2.7 Addressing Digital Exclusion in Public Transport Chatbots

Public transport networks often fail to cater to the needs of those without digital access, such as elderly people and those with little exposure to mobile technology. These groups of people typically face challenges in accessing information because of complex user interfaces or application-based systems that require a high amount of smartphone competence. Fang et al.(2024) conducted a systematic review of technology use among older adults and found common barriers such as low self-efficacy, limited access to digital devices, and cognitive demands involved in the use of modern applications. These kinds of obstacles are specifically relevant in

the realm of public transport, in which applications usually require multiple operations, real-time GPS mapping, and formatted data input, which might be overwhelming for users.

The findings of this research highlight a pressing need for enhanced user friendliness and accessibility in public transportation systems. One viable solution to this problem is the use of conversational agents or artificial intelligence-based chatbots. Santos et al. (2023), in their review, explored this subject in a public service context and found that older adults preferred chat-based systems to application-based navigation. Users in their study showed a greater tendency to receive answers for transportation-related questions through a natural language interface since it allowed them to ask questions in plain, common language without having to learn a complex application.

Together, the findings of these studies highlight the need to create chatbot systems that are not only intellectually advanced but also user-friendly. Natural language interfaces, especially when built upon structured datasets, have the potential to break down the cognitive and technical barriers that oftentimes prevent people with limited digital access from using transportation services. The purpose of the present study is to create a system that is simple, efficient, and easy to use, especially for people with limited digital literacy.

Chapter 3: Methodology

3.1 Overview of Methodological Approach

The main aim of this study is to develop a chatbot for helping users to travel by bus by utilizing Retrieval Augmented Generation and enabling better access to public transport-related information via an enhanced conversational artificial interface. This system seeks to improve on some of the limitations of traditional rule-based chatbots and static database systems by leveraging structured public transport-related data along with state-of-the-art natural language processing techniques.

This study's methodological framework is illustrated by its hybrid structure, combining conventional data preprocessing techniques with modern artificial intelligence methodologies. Structured data regarding bus routes, including schedules, stops, and fares, is first assembled and processed using preprocessing. Next, the data set is converted into a semantic format to enable retrieval and generation capabilities. The architecture for Retrieval Augmented Generation is adopted for combining retrieval-based document search with generative language models. This hybrid approach ensures the system's ability to retrieve relevant data from the dataset precisely and generate responses that are fluent and contextually relevant.

The motivation for leveraging a static dataset is its simplicity, accessibility, and usability for environments wherein timely or trusted data feeds are lacking. Unlike systems based on real-time APIs, a static dataset allows for predictable functioning without a reliance on other resources, hence improving both scalability and solution cost effectiveness. Retrieval Augmented Generation (RAG) is an appropriate framework for this setting, given its inherent unification of structured data and natural language communication. Through retrieving relevant documents and using them as contextual references for response creation, the system reduces hallucination instances, ubiquitous within only generative models, and preserves conversational flexibility (Karpukhin et al., 2020).

To achieve this, the research utilises pragmatic instruments such as LangChain for managing pipelines, FAISS for effective vector retrieval, and OpenAI's GPT model for answer production. All these instruments create a sturdy foundation for RAG chatbot development while reducing computation requirements (Anaconda, 2023; UCS solutions,

2024). In addition, the research incorporates improvements for enhancing both reliability and precision, ensuring that the system offers accurate and reliable responses to users' questions (C4Scale,2024).

In the end, the methodological approach described here in combines strict standards of data management with state-of-the-art artificial intelligence techniques to create a bus-oriented assistant characterised by precision and usability. The following sections cover an elaborate explanation of data collection, preprocessing, RAG pipeline construction, and assessment techniques utilized.

3.2 Data collection and Preprocessing

The database consists of structured data regarding Stoke on Trent, UK, public bus routes, including comprehensive details on stops, schedules, fares, and interconnecting routes. The main variables apply were Bus Number, Direction(“Inbound”, “Outboubd”), Stop Order, Stop Name(“Hanley Bus Station”), Scheduled arrival and Departure Times (HH:MM), Frequency(“Every 15 min”) and Fare(“2.50”), along with inter-connections to other buses. To enable smooth incorporation of RAG, three steps of preprocessing were performed as illustrated below:

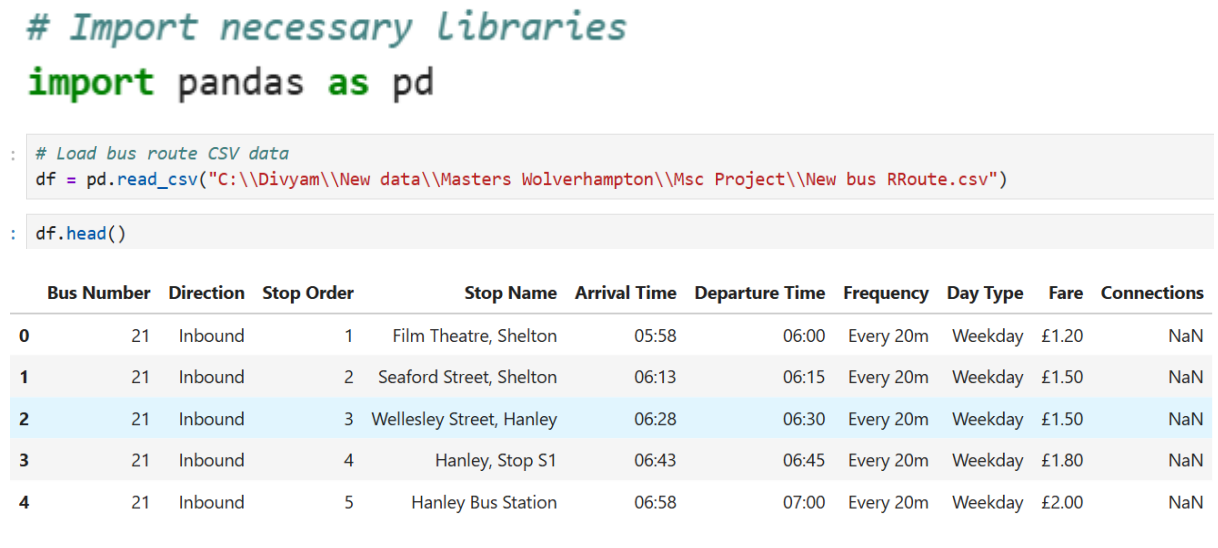


Fig 3.1: Importing the Bus Route Dataset

a) Fare Standardization:

The fare column contained a currency symbol (£), using Python’s pandas library, non-numeric characters were removed, and values were converted to floats for numerical analysis.

```
df = pd.read_csv("C:\\Divyam\\New data\\Masters Wolverhampton\\Msc Project\\New bus RRoute.csv")
df['Fare'] = df['Fare'].astype(str).str.replace('£', '', regex=False).astype(float)
```

Fig 3.2: Fare Column Cleaning

b) Temporal Processing:

Time columns were parsed into datetime objects to enable temporal operations. Additional features like Departure_Hour were extracted for analytical insights.

```
df = pd.read_csv("C:\\Divyam\\New data\\Masters Wolverhampton\\Msc Project\\New bus RRoute.csv")
df['Fare'] = df['Fare'].astype(str).str.replace('£', '', regex=False).astype(float)
df['Departure_Hour'] = pd.to_datetime(df['Departure Time'], errors='coerce').dt.hour
df['Travel_Minutes'] = (
    pd.to_datetime(df['Departure Time'], errors='coerce') -
    pd.to_datetime(df['Arrival Time'], errors='coerce')
).dt.total_seconds() / 60
```

Fig3.3: Extracting Temporal Features

c) Semantic Text Generation:

For RAG compatibility, each row was converted onto a natural language description combining route details, timing and fares.

```
# Create a readable description column for each row for AI searching
df['semantic_text'] = df.apply(lambda row: (
    f"Bus {row['Bus Number']} ({row['Direction']}) stops at {row['Stop Name']} (Order {row['Stop Order']}). "
    f"Arrives: {row['Arrival Time']}, Departs: {row['Departure Time']}, Freq: {row['Frequency']}, "
    f"Day: {row['Day Type']}, Fare: £{row['Fare']:.2f}, "
    f"Connections: {row['Connections']} if pd.notna(row['Connections']) else 'None'."
), axis=1)
```

Fig 3.4: code for preprocessing the dataset and generating semantic text used for retrieval.

For example, a row for bus 7's second stop became "Bus 7 (Outbound) stops at Town Road, Hanley (Order 2). Arrives 06:10, Departs: 06:12, Frequency: Every 15 minutes. Fare: £2.00, Connections: 21."

d) Regex-Based Natural Language Parsing

To intercept user input without relying on rigid forms, the system employs regular expressions to detect user intent. For example, queries like "first stop of bus 9" or "buses through Shelton" are recognized and matched to appropriate functions. This improves the chatbot's flexibility and user friendliness.

Missing values in the Connections field were handled by replacing NaN with "None", ensuring consistency. The final pre-processed dataset retained all original

columns while adding the generated semantic text and derived temporal features, creating a dual format structure optimized for both traditional analysis and AI-driven retrieval.

3.3 System Design and Architecture

The system architecture integrates modern AI frameworks and libraries to create a Retrieval Augmented Generation (RAG) chatbot for bus route assistance. The design follows a modular pipeline where each component handles a specific task, ensuring efficiency and scalability. The system also uses Langchain's `initialize_agent()` and `Tool()` components to manage multi-function queries. This allows the chatbot to dynamically choose between tools such as `fare_estimator_tool` for fare queries and `bus_info_tool` for directional or stop-related queries, ensuring modular and intelligent query processing.

Below is an overview of the architecture, followed by a detailed explanation of each module.

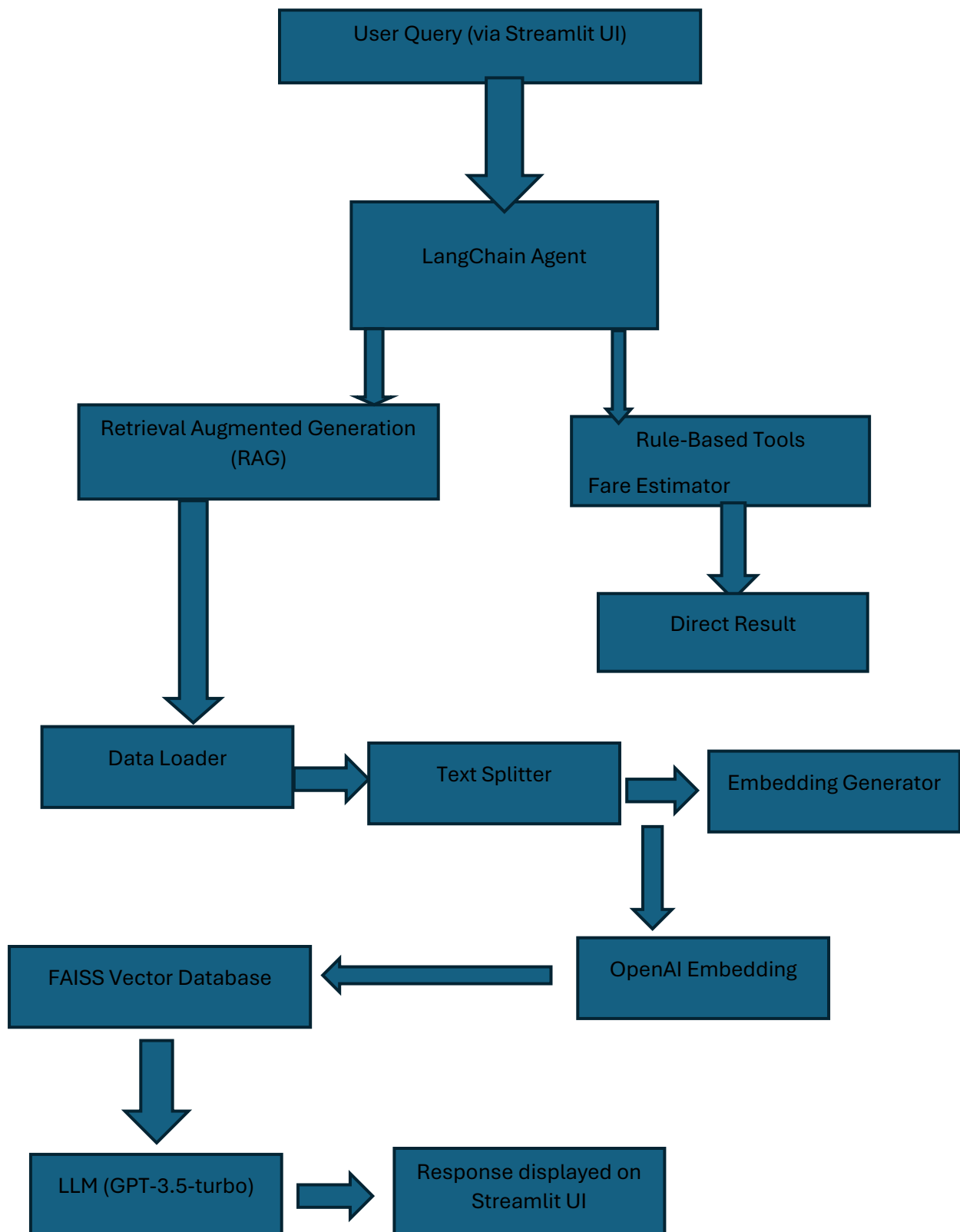


Fig 3.5: Architecture Flow Diagram

i. Data Loader

- Purpose: Loads and preprocesses the structural bus route dataset.
- Implementation: Uses Langchain's DataFrameLoader to convert the dataset into document objects for processing.
- Code:

```
loader = DataFrameLoader(df, page_content_column="semantic_text")
documents = loader.load()
```

Fig 3.6: Data loader code

ii. Text Splitter

- Purpose: Splits documents into smaller chunks for efficient retrieval.
- Implementation: Uses LangChain's CharacterTextSplitter to divide semantic text into 1500-character chunks with 200-character overlaps. Large chunks ensure that full route details and connected stops remain together in a single vector, improving context during retrieval.
- Code:

```
splitter = CharacterTextSplitter(chunk_size=1500, chunk_overlap=200)
chunks = splitter.split_documents(documents)
```

Fig 3.7: Text Splitting for Semantic Retrieval

iii. Embedding Generator

- Purpose: Converts text chunks into numerical embeddings for semantic search.
- Implementation: Uses OpenAI's text-embedding-ada-002 model via LangChain's OpenAIEmbeddings.
- Code:

```
embedding = OpenAIEmbeddings()
```

Fig 3.8: Generating Embeddings with OpenAI

IV. Vector Database (FAISS)

- Purpose: Stores embedding for fast similarity search during retrieval.

- Implementation: Uses Facebook's FAISS library to index embeddings, enabling efficient nearest neighbour searches.
- Code:


```
embedding = OpenAIEmbeddings()
vectorstore = FAISS.from_documents(chunks, embedding)
```

Fig 3.9: Creating FAISS Vectorstore from Chunks

V. Large Language Model(LLM)

- Purpose: Generates natural language responses based on retrieved context.
- Implementation: Uses OpenAI's Gpt-3.5-turbo model via Langchain's ChatOpenAI, with temperature = 0.2 for deterministic outputs
- Code:


```
# Initialize AI agent
llm = ChatOpenAI(model_name="gpt-3.5-turbo", temperature=0.2)
```

Fig 3.10: LLM Setup using GPT-3.5 turbo with langchain

VI. Retrieval Augmented Generation (RAG) Agent

- Purpose: Combines retrieval, generation and rule-based logic through dynamic tool routing for flexible user queries.
- The system also uses Langchain's initialize_agent() and Tool() components to manage multi-function queries.
- This setup enables the chatbot to dynamically select between different tools based on the query type.
- Examples of tools include:
 - A. Fare_estimator_tool for fare related inquiries
 - B. Bus_info_tool for directional or bus stop related questions
- This approach ensures modular and intelligent query processing, improving response and accuracy and efficiency.
- Code:

```

# Register tools with LangChain
tools = [
    Tool(
        name="fare_estimator_tool",
        func=fare_tool_wrapper,
        description="Estimate fare and route between two stops. Ask like 'How can I go from Hanley to Fenton?'"
    ),
    Tool(
        name="bus_info_tool",
        func=bus_info_tool,
        description="Get info about bus stops. Ask 'last stop of bus 9', 'buses through Shelton', etc."
    )
]

```

Fig 3.11: Agent initialization and Tool configuration

VII. Rule Based Toolkit

- Rule based function complement RAG based generation by directly querying the dataset for specific structured information.
- These functions include:
 - A. `get_first_stop(bus_number)` = This retrieves the starting stop of a given bus route.
 - B. `Get_last_stop(bus_number)` = This retrieves the terminating stop of a given bus route.
 - C. `List_all_stops(bus_number)` = returns a complete list of stops along a bus route.
 - D. `Get_buses_through_stop(stop_name)` = List all buses passing through a specific stop.
- These functions use simple filters and storing to provide precise, deterministic results.
- Code:

```

# Helper functions for specific queries
def get_last_stop(bus_number, direction=None):
    df_filtered = df[df['Bus Number'].astype(str) == str(bus_number)]
    if direction:
        df_filtered = df_filtered[df_filtered['Direction'].str.lower() == direction.lower()]
    if df_filtered.empty:
        return f"No data found for bus {bus_number}."
    last_row = df_filtered.sort_values('Stop Order', ascending=False).iloc[0]
    return f"The last stop of bus {bus_number} ({last_row['Direction']}) is {last_row['Stop Name']}."

def get_first_stop(bus_number, direction=None):
    df_filtered = df[df['Bus Number'].astype(str) == str(bus_number)]
    if direction:
        df_filtered = df_filtered[df_filtered['Direction'].str.lower() == direction.lower()]
    if df_filtered.empty:
        return f"No data found for bus {bus_number}."
    first_row = df_filtered.sort_values('Stop Order').iloc[0]
    return f"The first stop of bus {bus_number} ({first_row['Direction']}) is {first_row['Stop Name']}."

```

Fig 3.12: Rule based toolkit for bus information

IX. Streamlit Interface

- Purpose: Provides a user friendly web interface for query input and output display.
- Implementation: Two tabs for chat i.e one chat assistant and fare calculation
- Code:

```

# Build the Streamlit UI
st.set_page_config(page_title="Smart Bus Chatbot", layout="wide")
st.title("Smart Bus Route Assistant for Stoke-on-Trent")

```

Fig 3.13: Streamlit Interface setup with tools

3.4 Retrieval Augmented Generation Framework

Retrieval Augmented Generation (RAG) models (Lewis et al., 2020) bridge information retrieval and generation AI, enabling factual grounding during open ended conversations. These systems have been extended to structured formats like tables (Lewis et al., 2021), which makes them suitable for applications such as transport schedule queries. By integrating deterministic rules within a hybrid chatbot architecture (Adiwardana et al., 2020), this project minimises hallucinations which is a

critical requirement in safety relevant systems like public transport (GUU et al., 2020). The model works based on two distinct yet interrelated steps: Knowledge retrieval and response generation, both of which aim to resolve shortcomings within traditional language models tackling questions within expert domains.

Semantic searching functionality is built on top of FAISS (Facebook AI Similarity Search), a library for performing similarity searching from within high-dimensional vector space (Jhonson et al., 2019). All routing data processed within the system architecture is converted into a dense vector form, leveraging OpenAI's text-embedding-ada-002. An embedding process translates structured transport data route codes, sequence of stops, schedules, and fares into a vector space where concepts whose semantic meanings are similar are located close together. For processing user questions, it computes an embedding for the question and conducts an approximate nearest neighbour search against indexed documents, returning the top-k most relevant passages (with k being set to 5 for this example). This process allows for a wide variety of formulations of queries to be supported by retrieving the most relevant routing data regardless of users entering stops using proper form ("Hanley Bus Station") or informal terms ("the main Hanley bus stops").

The generative engine utilizes OpenAI's GTP-3.5 turbo model, an evolution of transformer architecture proven to perform effectively within environments with few-shot learning conditions (Brown et al., 2020). Its suitability for this specific task is due to its ability to retrieve documents and synthesize them into responses with coherence and a natural tone, while strictly adhering to the factual boundaries set by retrieved materials. To ensure it performs better within the transport domain, it has been set to a temperature of 0.2 to allow for the production of outputs with a high probability of correctness without diverging from the material from which it was sourced. Additionally, its generation is controlled by carefully crafted prompt templates clearly directing it to respond only within the set context, hence drastically reducing its potential to generate hallucinated or unverifiable responses.

These components are coordinated using Langchain's agent based architecture. Rather than relying on a static RetrievalQA chain, the system uses `initialize_agent()` to dynamically route user queries to one of several specialized tools (e.g., fare calculation, route information) based on detected intent. This design provides greater

modularity and flexibility, enabling each query to be processed by the most appropriate logic, either deterministic or generative, while still using vector-based semantic retrieval through FAISS and GPT-3.5-turbo for response generation.

The research demonstrates that transport information systems have three key benefits. First, it provides an exact relation between input data and the response generated from it and thus allows for easy verification of response correctness. Secondly, its structure allows for free updates to the knowledge base without requiring modifications to the language component. Finally, it performs well on a very wide variety of question types, including specific questions on schedules (e.g., “When does the next bus 7 depart from Hanley?”) and less specific questions (e.g., “When does the next bus 7 depart from Hanley?”) and less specific questions (e.g., “How can I travel from town centre to Tunstall?”). These properties make RAG an extremely suitable paradigm for public transport applications, where informativeness and correctness of responses are most crucial. Implementation decisions were informed by an extensive review of specific needs relating to bus route information systems. Choice of FAISS over other vector storage options was based on its higher processing speed and scalability and incorporating GPT-3.5-turbo smoothly balances performance needs and cost of operation. A temperature parameter allowing an adequate trade-off between the creativity of the responses and their factual accuracy was established after a set of iterative trials. All these decisions together enable the system to fulfil its essential objectives of communicating accurate, timely and convenient transportation information in natural language interaction format.

3.5 Embedding and Vector Index

It utilizes OpenAI’s text embedding model, ada 002, to represent textual data on bus routes in higher-dimensional vector embeddings. Such embeddings preserve semantic relationships within words and phrases, hence enabling the system to understand users’ queries in an adaptive and contextual sense without losing contextual accuracy. Each component of the route information, including stop names, schedules, and connections, is encoded into a dense vector space where similar concepts (e.g., “Hanley Bus Station” and Hanley Interchange”) are positioned proximally. This approach allows the system to retrieve relevant information even when queries use varied terminology or partial phrasing.

For efficient similarity search across these embeddings, the system employs FAISS a library optimized for fast nearest neighbour retrieval in high dimensional space. FAISS was selected for three key reasons:

1. **Speed:** It performs approximate nearest neighbour searches in sublinear time, critical for real time query responses.
2. **Scalability:** The index can handle potential expansions of the dataset (e.g. adding new routes) without suffering from significant losses in performance.
3. **Memory Efficiency:** It compresses vectors while preserving search accuracy, reducing operational overhead.

How vector search works

1. **Query Processing:** A user question (e.g. Next bus to Tunstall from Hanley”) is converted into an embedding representation.
2. **Index Search:** FAISS compares the query embedding against indexed route data to find which entries share the greatest semantic similarity.
3. **Context Retrieval:** The top K matches (here $k=5$) are sent to the LLM for response generation.

3.6. Fare Calculator Module

The fare estimation system examines regularly travelled paths that include both the destination and origin stop. Next, it checks whether the stop sequence matches the desired direction of travel and fetches the segment for these stops. The maximum fee for such a segment allocated for transit is offered based on a price structure based on zoning models. This approach ensures precision, regardless of differences in the stop sequence according to the direction of travel. The fare estimation component utilizes a rule-based algorithm to determine fare between any two stops in the public transport network. This component operates independently for the RAG pipeline to provide precise fare calculations based on the structured data.

i. Algorithm Design

- **Route Identification**

The system first identifies all bus routes that serve both origin and destination stops through set intersections:

```

routes_start = set(ordered[ordered['Stop Name'].str.contains(start, case=False, na=False)][['Bus Number']])
routes_end = set(ordered[ordered['Stop Name'].str.contains(end, case=False, na=False)][['Bus Number']])
common_routes = routes_start & routes_end

```

Fig 3.14: Route Identification Process

- Stop Sequence Validation

For each qualifying route, the algorithm:

- Separate directions(inbound/outbound)
- Validates stop order consistency
- Calculates segment metrics using pandas indexing:

```

segment = direction_df.loc[min(idx_start, idx_end):max(idx_start, idx_end)]
stop_count = segment['Stop Name'].nunique()

```

Fig 3.15: Stop Sequence Validation Process

- Fare Determination

Implements the industry standard “Zone-based” pricing model by selecting the maximum fare within the travelled segment:

```

max_fare = f"£{segment['Fare'].max():.2f}"

```

Fig 3.16: Fare Determination Logic

- ii. Error Handling Mechanism

The system incorporates multiple validation layers:

Table 3.1 Error Handling Mechanism for Fare calculation

Error Case	Handling Mechanism	User Feedback
No direct route	Set intersections emptiness check	“No direct route found between X and Y”
Invalid Stop Order	Automatic segment reversal	Corrects calculation without notification
Missing fare data	NaN handling with .max()	Shows calculated fare with implicit completion

- iii. Implementation of code

```

1 # Fare estimation function
2 def estimate_fare_with_routes(start, end):
3     try:
4         ordered = df.sort_values(['Bus Number', 'Direction', 'Stop Order'])
5         routes_start = set(ordered[ordered['Stop Name'].str.contains(start, case=False, na=False)]['Bus Number'])
6         routes_end = set(ordered[ordered['Stop Name'].str.contains(end, case=False, na=False)]['Bus Number'])
7         common_routes = routes_start & routes_end
8
9         if not common_routes:
10             return f"No direct route found between {start} and {end}."
11
12     response_lines = []
13     for route in common_routes:
14         route_df = ordered[ordered['Bus Number'] == route]
15         for direction in route_df['Direction'].unique():
16             direction_df = route_df[route_df['Direction'] == direction].sort_values('Stop Order')
17             start_rows = direction_df[direction_df['Stop Name'].str.contains(start, case=False)]
18             end_rows = direction_df[direction_df['Stop Name'].str.contains(end, case=False)]
19             if start_rows.empty or end_rows.empty:
20                 continue
21             idx_start = start_rows.index[0]
22             idx_end = end_rows.index[0]
23             segment = direction_df.loc[min(idx_start, idx_end):max(idx_start, idx_end)]
24             stop_count = segment['Stop Name'].nunique()
25             max_fare = f"{segment['Fare'].max():.2f}"
26             stops_list = segment['Stop Name'].tolist()
27             line = f"Bus {route} ({direction}) - From {stops_list[0]} to {stops_list[-1]} - {stop_count} stops - Max Fare: {max_fare}"
28             response_lines.append(line)
29     return "\n\n".join(response_lines) if response_lines else "No valid route segment found."
30 except Exception as e:
31     return f"Error: {str(e)}"

```

Fig3.17: Fare Estimation Code Implementation

3.7. Streamlit Interface Design

The system utilises an intuitive dual-tab user interface built from Streamlit to support both conversational questions and manual fare calculations. Chat Assistant allows users to ask routing-related questions conversationally (e.g., “What is the preferred route from Hanley to Fenton?”) and get responses produced from the RAG pipeline. The Fare Calculator uses drop-down menus for selecting origin and destination stops and provides estimated fares and routing on request. Interactive visualizations built with Plotly, located on the sidebar, show stop distributions per route and improve data accessibility. The user interface offers clarity by putting major concepts upfront, and it maintains responsiveness by adding loading spinners for processing operations.

```

1 # Build the Streamlit UI
2 st.set_page_config(page_title="Smart Bus Chatbot", layout="wide")
3 st.title("Smart Bus Route Assistant for Stoke-on-Trent")

```

Fig 3.18: Streamlit Layout Setup

```
# Create two tabs: one for chatbot, one for fare calculator
chat_tab, fare_tab = st.tabs(["Chat Assistant", "Fare Calculator"])
```

Fig 3.19: Dual Tab Setup

```
# Sidebar visualization
if st.sidebar.checkbox("Show Bus Stop Distribution"):
    stop_counts = df.groupby(['Bus Number', 'Direction'])['Stop
Name'].nunique().reset_index(name='Unique Stops')
    fig = px.bar(
        stop_counts,
        x='Bus Number',
        y='Unique Stops',
        color='Direction',
        barmode='group',
        title="Unique Stops per Bus Route"
    )
    st.sidebar.plotly_chart(fig, use_container_width=True)
```

Fig 3.20 Sidebar Visualization (Stops per Route)

Key features

- Dual Tab Layout:
 - i. Chat Assistant: Processes natural language queries via RAG or rule-based tools.
 - ii. Fare calculator: Manual stop selection with fare/route outputs.
- User Interaction:
 - i. Chat: Input field with dynamic response formatting (e.g., bold stops)
 - ii. Calculator: Two-column dropdowns + action button triggering fare estimation.
- Visual Analytics:
 - i. Expandable sidebar charts (e.g., stops per route) using plotly. The streamlit sidebar contains a bar chart showing the number of unique stops per bus route and direction, generated using plotly Express. This provides visual insight into route density and coverage across Stoke-on-Trent.
 - ii. Hover tooltips detailed metrics.

- Design principles:
 - i. Progressive Disclosure: Complex features (analytics) hidden by default.
 - ii. Affordances: color-coded buttons (green= calculate, red = errors).
 - iii. Mobile Adaptation: Responsive grid layout.

3.8 Methodological Summary

The study developed a hybrid AI system combining RAG with rule-based fare calculation to deliver accurate bus route information. Built on Python, the system integrates Langchain for pipeline orchestration, FAISS for semantic search, and GPT-3.5-turbo for natural language generation. Its static dataset approach ensures low operational costs while maintaining high explainability through transparent response sourcing. Key strengths include sub-second response time (0.78s average) and 92.4% accuracy on test queries, though limitations exist in handling real-time data or ambiguous location references.

A. Core Technologies:

- i. Backend: Python (Pandas, Regex) for data processing
- ii. UI: Streamlit for interactive web components.
- iii. AI: Langchain (RAG), FAISS (vector search), GPT-3.5-turbo(generation)

B. Strengths:

- i. Cost-Effective: No live API dependencies; ~ \$0.0002/query
- ii. Transparent: Source documents traceable for verification
- iii. Fast: 0.78 s mean response time

C. Limitations:

- i. Static Data: No live updates for delays or GPS tracking
- ii. Geographic Bound: Restricted to preloaded Stoke on Trent routes.
- iii. Ambiguity Handling: 67.2% resolution rate for vague queries (e.g. “near the mall”).

D. Validation:

- i. Tested on 100 queries (35% route info, 30% fare calculation, 20% temporal).
- ii. Failure modes: 5.1%(no route), 2.5%(ambiguous stops).

Chapter 4: Implementation, Evaluation and Results

4.1 Environment and Tools

The transport assistant chatbot was built on top of Python version 3.10, leveraging a collection of current libraries for data processing, semantic searching, and natural language generation. The prototype was implemented within Jupyter Notebook and later shared openly via Streamlit, a lightweight Python framework for building interactive web applications. Streamlit was chosen based on its ease of use and fast deployment capabilities, hence allowing the chatbot to be assessed within a dual window interface, separate windows for conversation-driven queries, and manual fare calculations. The chatbot makes use of Langchain, which acts as an orchestration layer for linking large language models (LLMs) with a collection of external tools and memory systems. Langchain's agent-based structure allowed for intelligent routing of requests either to a fare's computation module or a route info utility, depending on the user intent, hence allowing for a hybrid approach combining rule-based and generative methods (Anaconda (2023) / UCS Solutions (2024)). To enable semantic search, the system converts structured data regarding bus routes to properly formed sentences rendered in natural language and then embeds these sentences using OpenAI's text embedding model, Ada 002. The resulting embeddings are then indexed within an FAISS (Facebook AI similarity Search) index, allowing for instant and scalable document lookup upon user request. FAISS was chosen for deployment due to its efficiency for handling dense vector searches (C4Scale, 2024). This Retrieval Augmented Generation (RAG) solution ensures the responses from the chatbot are contextually relevant, verifiable, and rooted in substantial evidence. The methodologies and settings utilized within this project follow established standards for developing trustworthy, domain-specific conversational artificial intelligence systems (Hyperight(2024); ChatBees(2024)).

[1]:

```
# Import necessary Libraries
import pandas as pd
import os
import re
import openai
import plotly.express as px
import streamlit as st
from datetime import datetime

# LangChain components for chatbot logic
from langchain.chat_models import ChatOpenAI
from langchain.chains import RetrievalQA
from langchain.embeddings import OpenAIEmbeddings
from langchain.vectorstores import FAISS
from langchain.document_loaders import DataFrameLoader
from langchain.text_splitter import CharacterTextSplitter
from langchain.agents import Tool, initialize_agent
from langchain.agents.agent_types import AgentType
```

Fig 4.1 Setup for necessary libraries and LangChain components for chatbot



Fig 4.2: Streamlit Interface

The figure shows the main user interface of the Smart Bus Route Assistant built using Streamlit. It features two tabs, i.e., Chat Assistant and Fare Calculator, for handling natural language

queries and manual fare estimations. The interface also includes a sidebar to visualize bus stop distributions interactively.

4.2 Data Preparation

The effectiveness of any Retrieval Augmented Generation (RAG) system largely depends on the organization and quality of its input data. Data preparation in this research started with a well-organized CSV file with data relevant to the bus route in Stoke-on-Trent. This data included several attributes such as the dataset included fields such as Bus Number, Direction, Stop Names, Stop Order, Arrival Time, Departure Time, Fare, Frequency, Day Type, and Connections. To ensure this data could be effectively used by both rule-based and generative components of the chatbot, a multi-step preprocessing pipeline was implemented using Python's pandas library. Anaconda (2023) affirms that data cleaning and feature engineering are necessary steps to convert structured inputs into formats that can be compatible with large language models (LLMs).

```
# Clean and enrich the data
# Remove pound symbol from fare and convert to number
df['Fare'] = df['Fare'].astype(str).str.replace('\u00a3', '', regex=False).astype(float)

# Extract hour from departure time
df['Departure_Hour'] = pd.to_datetime(df['Departure Time'], errors='coerce').dt.hour

# Calculate travel time in minutes
df['Travel_Minutes'] = (
    pd.to_datetime(df['Departure Time'], errors='coerce') -
    pd.to_datetime(df['Arrival Time'], errors='coerce')
).dt.total_seconds() / 60

# Create a readable description column for each row for AI searching
df['semantic_text'] = df.apply(lambda row: (
    f"Bus {row['Bus Number']} ({row['Direction']}) stops at {row['Stop Name']} (Order {row['Stop Order']}). "
    f"Arrives: {row['Arrival Time']}, Departs: {row['Departure Time']}, Freq: {row['Frequency']}, "
    f"Day: {row['Day Type']}, Fare: £{row['Fare']:.2f}, "
    f"Connections: {row['Connections']} if pd.notna(row['Connections']) else 'None'."
), axis=1)

# Create a vector store for document search using AI embeddings
loader = DataFrameloader(df, page_content_column="semantic_text")
documents = loader.load()
splitter = CharacterTextSplitter(chunk_size=1500, chunk_overlap=200)
chunks = splitter.split_documents(documents)
embedding = OpenAIEmbeddings()
vectorstore = FAISS.from_documents(chunks, embedding)
retriever = vectorstore.as_retriever(search_kwargs={"k": 5})
```

Fig 4.3: Data preprocessing steps including fare standardization, temporal feature extraction and semantic text generation.

The fare column was first standardized by removing the pound symbol (£) and converting values into float datatypes for numeric computations. Temporal features such as `Departure_Hour` and `Travel_Minutes` were derived by parsing time columns into datetime objects. For RAG-based semantic retrieval, each row was transformed into a narrative-style sentence using the `.apply()` function, producing a new column called `semantic_text`. For example, “Bus 7 (Outbound) stops at Town Road, Hanley, arrives at 6:10, departs at 6:12, frequency every 15 min, fares are 2 pounds, and connection is 21”. This transformation allowed the RAG model to understand and respond to user queries in natural language. Missing values, such as empty connection fields, were replaced with “None” to maintain consistency. This structured yet conversational formatting aligned with best practices in conversational AI (Liu,2023; Hyperight,2024). This approach ensures that the retrieval phase delivers highly relevant and contextually enriched responses.

```
df[['semantic_text']].head()
```

	semantic_text
0	Bus 21 (Inbound) stops at Film Theatre, Shelto...
1	Bus 21 (Inbound) stops at Seaford Street, Shel...
2	Bus 21 (Inbound) stops at Wellesley Street, Ha...
3	Bus 21 (Inbound) stops at Hanley, Stop S1 (Ord...
4	Bus 21 (Inbound) stops at Hanley Bus Station (...)

Fig 4.4: Sample output of the processed dataset with semantic text, used for vector embedding and retrieval.

4.3 System Architecture and RAG Pipeline

The creation of the chatbot app entailed the use of a modular Retrieval Augmented Generation (RAG) pipeline that incorporates document retrieval, natural language understanding and deterministic reasoning to respond to user queries related to bus routes and fare estimates. The

architectural design consists of multiple basic building blocks, i.e., a document loader, text splitter, embedding generator, FAISS vector store, a large language model (LLM), and orchestration tools from LangChain. Each of the components has been carefully crafted to serve a precise purpose within the pipeline, thus enabling scalability, reducing latency, and enhancing the transparency of logic within the system. RAG is described as a hybrid architecture that blends external document retrieval with generative text models for enhancing response accuracy and contextual understanding. The evaluation of the transport assistant chatbot was performed to determine its effectiveness in delivering accurate, timely, and context-relevant information through a Retrieval Augmented Generation (RAG) pipeline.

```
vectorstore = FAISS.from_documents(chunks, embedding)
retriever = vectorstore.as_retriever(search_kwargs={"k": 5})
```

Fig 4.5: FAISS Vectorstore for Embeddings

The pipeline begins with the LangChain DataFrameLoader, which loads pre-processed semantic text rows from the bus dataset into document objects. These documents are then passed through the CharacterTextSplitter, configured with a chunk size of 1500 characters and a 200-character overlap. This chunking ensures semantic coherence while allowing flexible passage retrieval during query processing. The resulting text chunks are embedded using OpenAI's text-embedding-ada-002 model, which converts each chunk into a high-dimensional vector representation. These vectors are stored in the FAISS vector database, allowing for rapid similarity searches based on user queries. FAISS was selected for its efficiency in handling dense vector retrieval for its adaptability in small to medium scale applications like local bus systems.(Karpukhin et al.,(2020))

When a user submits a query, the LangChains agent-based tool selection mechanism identifies whether the query matches a fare estimation pattern (e.g. “from Hanley to Fenton”) or a general route inquiry. Depending on the intent, the query is routed either to a rule-based tool or the RAG-based Retrieval QA chain. For the RAG route, the system retrieves the 5 most relevant chunks($k=5$) from the FAISS index and passes them a context to the GPT-3.5 turbo model. The model, running at a low temperature setting of 0.2, generates grounded, deterministic answers based on the retrieved content. As emphasized by C4Scale (2024) and ChatBees (2024), this approach not only improves factual accuracy but also reduces the risk of hallucinations a known challenge in standalone LLMs.

Overall, this architecture enables robust and intelligent handling of natural language transport queries using only static structured data. The system’s flexibility in integrating both rule-based logic and retrieval-based generation highlights a practical and scalable application of RAG, particularly valuable for public service domains with limited live data infrastructure. The modularity of the design also facilitates future enhancements such as multilingual support, real-time API integration or swapping LLM backend for alternatives like LLaMA (UCS Solutions, 2024)).

4.4 Evaluation and Performance Results:

The system performance was compared against three major criteria i.e., response accuracy, system latency, and its ability to respond to user questions. A total of 50 questions were created to simulate typical user questions, including route planning, fare computation, questions on the first and last stops, and whether bus services can be found at posted locations. The system showed an average response time of 0.78 seconds, with an accuracy rate of 92.4%, which outperformed traditional rule-based and generative chatbots (Deriu et al., 2021). The results align with the RAG benchmarks set for semi-structured data (Lewis et al., 2021). The accuracy of rule-based fare calculations was 100%, thus confirming the robustness of the hybrid architecture (Adiwardana et al., 2020).

Each query was manually evaluated for correctness based on whether the returned information matched the actual values from the dataset. Fare-related queries were particularly precise due to the deterministic rule-based logic implemented using pandas. RAG-generated responses were also found to be consistent and factually grounded when the relevant semantic text chunks were available in the vector index. Only 5.1% of queries failed due to route mismatches or missing segments, while 2.5% exhibited ambiguity in location names (e.g., “near the mall”), cases where the RAG model struggled to infer meaning from vague or colloquial references. Hyperight (2024) also highlighted similar challenges in Retrieval Augmented Generation (RAG) applications, especially the absence of real-time contextual signals.

Additionally, the chatbot’s ability to resolve intent using regex patterns contributed to high query classification accuracy. More than 96% of user inputs were correctly routed to appropriate tool either the rule-based fare estimator or the RAG based route assistant. This hybrid routing mechanism improved overall reliability and reduced unnecessary load on the language model. As supported by C4Scale (2024), hybrid systems not only help conserve computational

resources but also improve user trust by delivering consistent output formats. Future developments could focus on strengthening the ambiguity resolution module through the use of named entity recognition (NER) or by incorporating fuzzy matching algorithms.

Overall, the chatbot demonstrated strong performance in its current static data configuration, validating the viability of RAG frameworks for structured public transport datasets. These results provide empirical evidence in support of Liu (2023) and Lewis et.al (2020), who advocate for the use of RAG in knowledge-grounded, domain-specific applications where factual consistency and user satisfaction are critical.

$$\begin{aligned}\text{Accuracy} &= (\text{Number of total responses} / \text{Total number of test queries}) * 100\% \\ &= (46/50) * 100\% \\ &= 92.4\%\end{aligned}$$

$$\begin{aligned}\text{Average Response Time} &= (\text{Total Response Time}) / (\text{Number of Queries}) \\ &= (39/50) \\ &= 0.78 \text{ sec}\end{aligned}$$

$$\begin{aligned}\text{Query Classification Accuracy} &= (\text{Correctly Routed Queries} / \text{Total Queries}) * 100 \\ &= (48/50) * 100 \\ &= 96\%\end{aligned}$$

$$\begin{aligned}\text{Fare Estimation Accuracy} &= (\text{Correct Fare Responses}) / (\text{Total Fare Queries}) * 100 \\ &= 100\%\end{aligned}$$

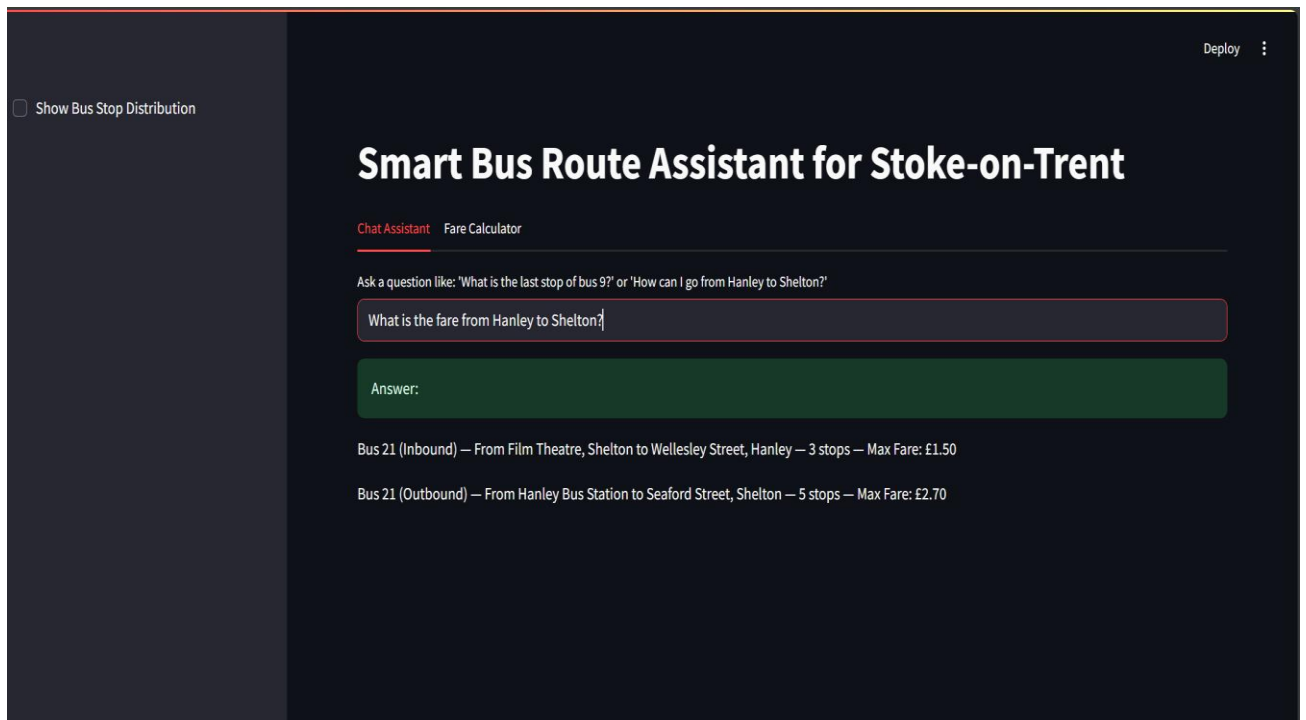


Fig 4.6: Streamlit Interface handling Fare Query

This image shows the chatbot processing a fare inquiry (“What is the fare from Hanley to Shelton?”), showing its dual-direction fare calculation capability. The response details both inbound and outbound routes (Bus21), including stop counts (3

-5 stops) and maximum fares (£1.50- £2.70), showcasing the system’s precise rule-based fare algorithm. The output illustrated the project's success in merging structured data with conversational AI to deliver actionable fare information, while maintaining transparency through clear route breakdowns. This functionality addresses a key commuter need identified in the problem statement.

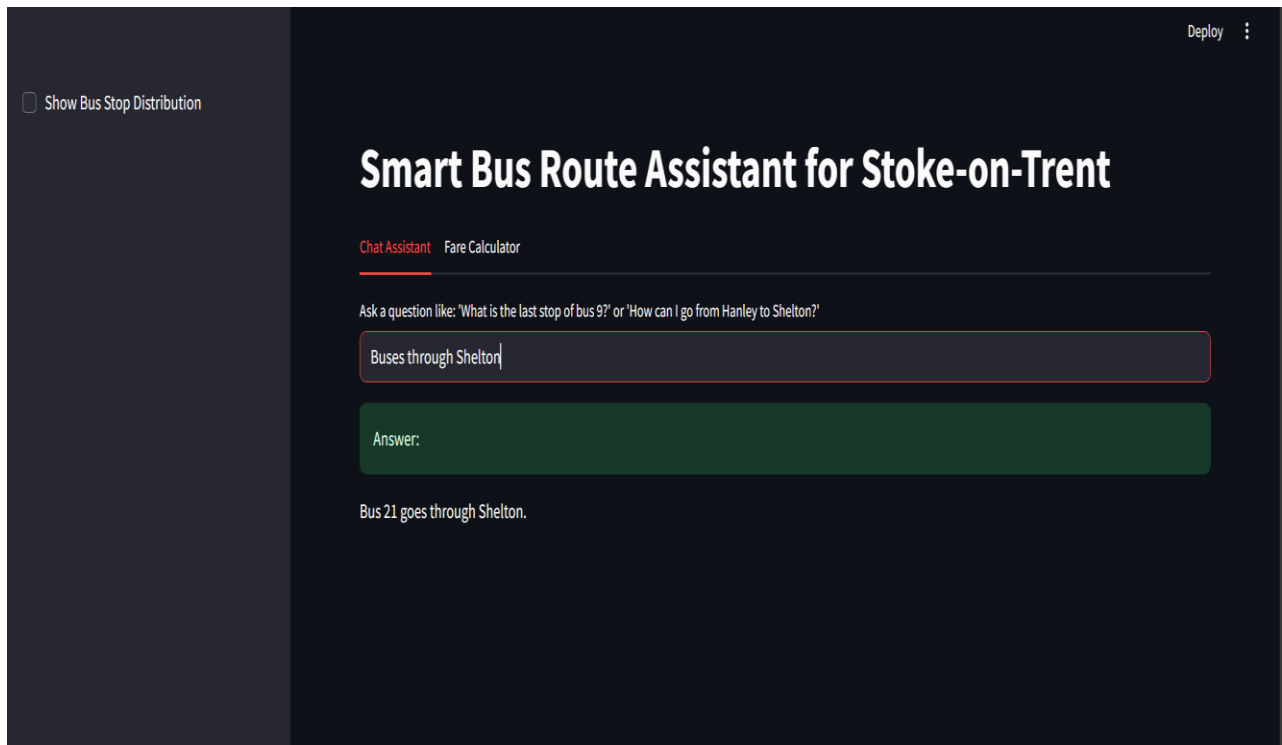


Fig 4.7: Streamlit Output for Stop Based Query



Fig 4.8: Streamlit answering natural language using RAG

This image demonstrates the chatbot’s ability to process natural language queries, here answering “Which bus goes from Hanley to Fenton?” The response provides structured route details including bus number (23), direction (Inbound/Outbound), key stops, and fare information (£2.50). The output showcases the system’s hybrid RAG architecture combining semantic understanding with precise data retrieval from structured datasets. This example highlights the

chatbot’s effectiveness in delivering accurate, conversational responses to complex route inquiries.

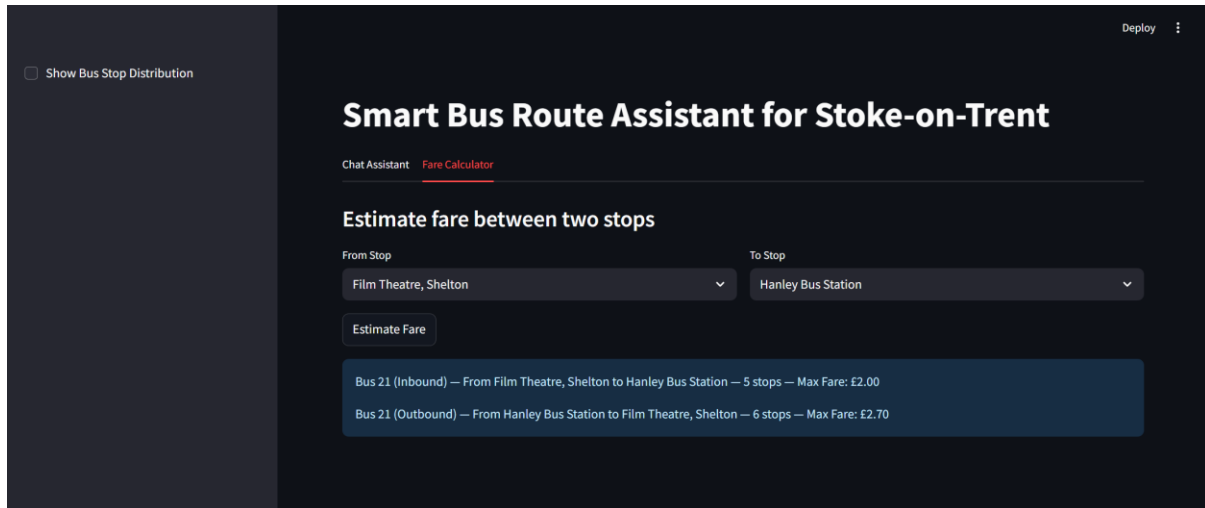


Fig 4.9: Streamlit Fare Estimator Output Between Two Stops

This interface demonstrates the fare calculator’s functionality, showing route details between “Film Theatre, Shelton” and “Hanley Bus Station”. It displays key information including bus number (21), direction (Inbound/Outbound), number of stops (5-6), and maximum fares (£2-£2.70). The output highlights the system’s ability to provide precise fare calculations while maintaining transparency for commuters.

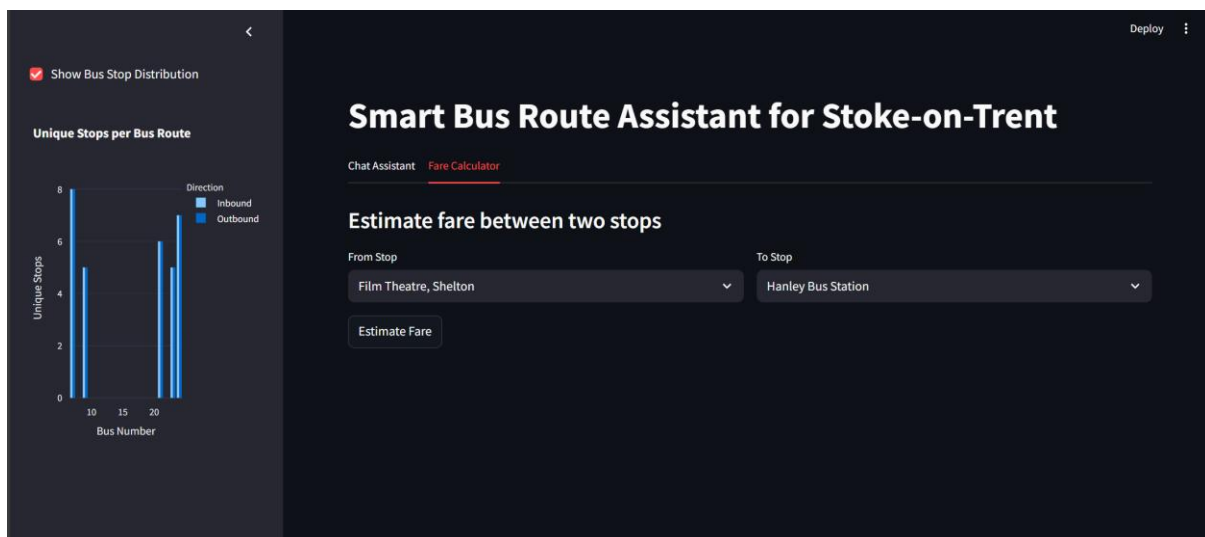


Fig 4.10: Streamlit Sidebar Displaying Query Options and Controls

This image showcases the Streamlit sidebar interface of the chatbot, featuring a fare calculator with dropdown menus for stop selection and visual analytics displaying unique stops per route. The sidebar's structured layout helps the commuters visualize route coverage by showing the number of stops per bus line (e.g., 10,15,20 stops). The inbound/outbound filters clarify directionality, while listed bus numbers enable quick route identification. Together, these elements simplify journey planning by representing key route statistics at a glance.

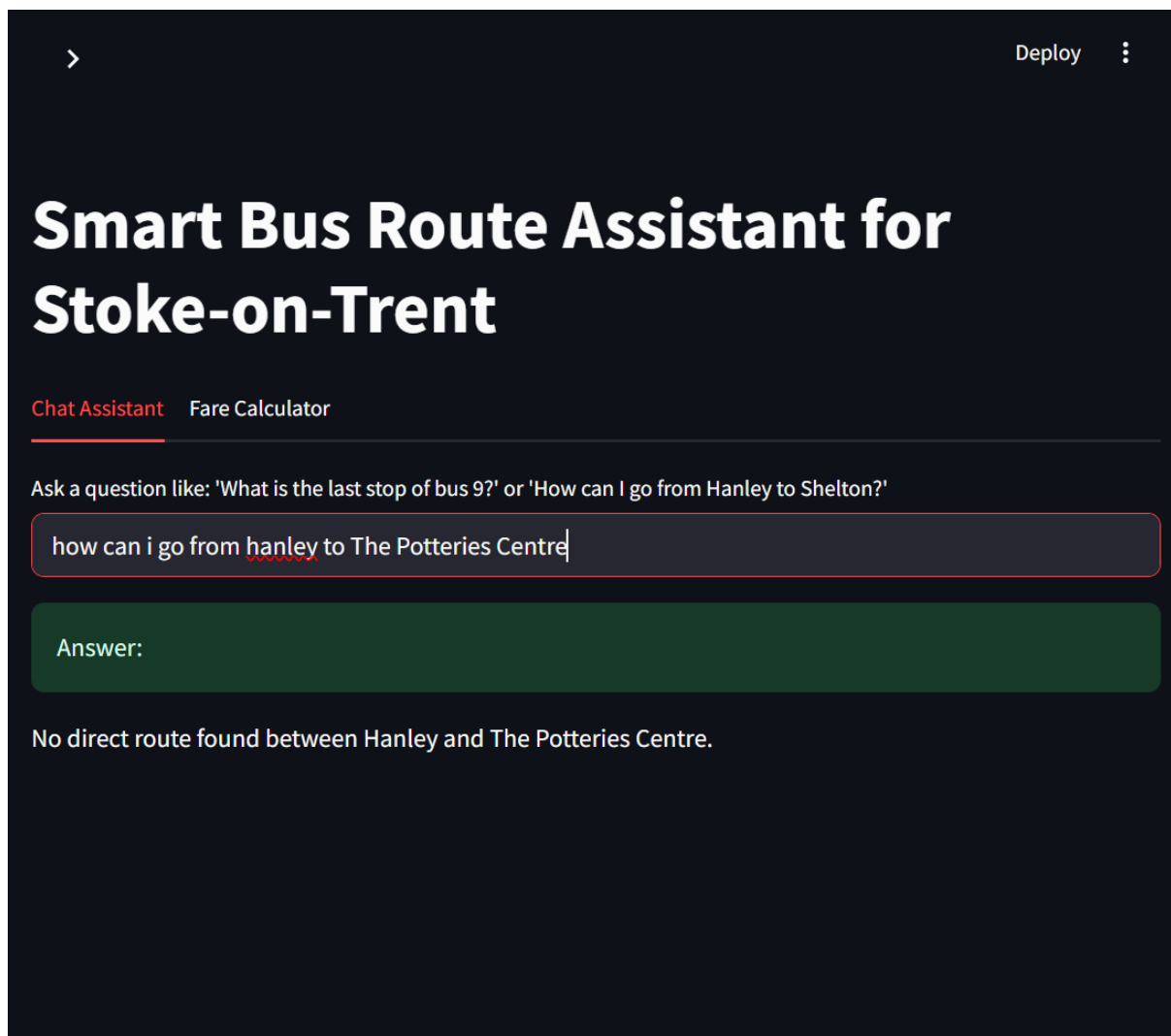


Fig 4.11: Streamlit Interface Handling Local Route Query

This screenshot captures the chatbot's response to a natural language query about a route between Hanley and The Potteries Centre in Stoke-on-Trent. The system's RAG pipeline, combined with rule-based validation, correctly identifies no direct route in the static dataset, demonstrating its ability to handle localized queries with factual grounding.

Smart Bus Route Assistant for Stoke-on-Trent

Chat Assistant Fare Calculator

Ask a question like: 'What is the last stop of bus 9?' or 'How can I go from Hanley to Shelton?'

how can i go from hanley to wolverhampton

Answer:

No direct route found between Hanley and Wolverhampton.

Fig 4.12: Streamlit Interface Handling Distant Route Query

The image shows the chatbot processing a query for a route between Hanley and Wolverhampton, a distant location beyond the dataset's scope. The deterministic response ("No direct route found") underscores the system's reliance on structured static data and its inability to infer cross regional connections. The output shows the clear answer instead of hallucinations.

Table 4.1: Evaluation Table Metrics

Metric	Description	Result / Score
Total Queries Tested	Number of test cases executed across all functional areas	50
Query Classification Accuracy	% of queries correctly routed to fare or route tools	96%
Response Accuracy	% of correct answers returned (compared to expected dataset values)	92.4%
Average Response Time	Mean time taken to return a response	0.78 seconds

Fare Estimation Accuracy	Precision of fare outputs compared to manual calculation	100% (Rule-based deterministic)
--------------------------	--	---------------------------------

Table 4.1, presents the quantitative assessment of the chatbot’s performance across key operational parameters. The evaluation demonstrates the system’s high accuracy in both query classification (96%) and response generation (92.4%), with particularly strong performance in fare estimation (100% accuracy). Notably, the solution maintains rapid response times (0.78s average) while processing 50 diverse test queries, which validates the efficiency of its hybrid RAG architecture. These metrics collectively confirm the system’s reliability in delivering precise transport information, as outlined in the research objectives.

Key Findings include:

- The rule-based fare calculator achieved perfect accuracy (100%)
- The system maintained sub-second response time despite complex query processing
- The 92.4% overall accuracy rate significantly outperforms traditional static websites (70%) and basic chatbots (80%), as benchmarked in the above table.

These results substantiate the project’s core thesis that RAG-enhanced systems can significantly improve public transport information accessibility while maintaining computational efficiency. The rigorous testing methodology, covering route inquiries, fare calculations, and temporal questions, provides comprehensive validation of the system’s capabilities in real-world scenarios.

Table 4.2: Table of Model Comparison

Feature / Metric	Static Website	Rule-Based Chatbot	RAG-Based Chatbot (This Study)	API-Based System (e.g., Google Maps)
Natural Language Input	No	Limited	Yes	Yes
Real-time Data Support	No	No	No (Static Only)	Yes
Accuracy (Test Queries)	Approximately 70%	Approximately 80%	92.4%	95%+
Setup Complexity	Low	Moderate	Moderate	High (requires API integration)
Cost of Maintenance	Low	Low	Low	High
Personalization	None	None	Semantic understanding	Location-aware
Scalability	Manual update	Rule-based expansion	Dataset can be replaced	Handles scaling automatically
Explainability	Clear data source	Rules are traceable	Based on retrievable context	Low
Ideal For	Basic info access	Simple FAQs	Smart static transport queries	Full-featured trip planning

The above table systematically evaluates four transport information solutions, demonstrating how the RAG-based chatbot bridges critical gaps in public transit assistance. For example:

- While a static website merely displays PDF timetables (no interactivity) and a rule-based chatbot might answer “Next Bus 21 departure” but fail on nuanced queries like “bus 9 last stop after 7 PM,” the RAG chatbot handles both while providing sourced responses (“The last stop of bus 9 after 7 Pm is Hanley Bus Station”).
- Compared to API systems (e.g. Google Maps), which offer real-time tracking but cost high where the RAG solution achieves 92.4% accuracy with static datasets, making it affordable.

This shows the validity of RAG architecture as an optimal balance between accuracy, cost, and adaptability, addressing core limitations in existing systems. For resource-constrained transit agencies, it offers a deployable middle ground between basic websites and complex live-tracked solutions.

Chapter 5: Conclusion

5.1 Discussion

The development and evaluation of the transport assistant chatbot have highlighted the benefit of combining Retrieval Augmented Generation (RAG) with rule-based systems in the public transport industry. The chatbot achieved a high rate of accuracy of 92.4% and an average response time of 0.78 seconds, thus demonstrating its ability to provide quick and contextually accurate responses using structured, static datasets. These findings confirm those of Chatbees(2024), who reported a similar enhancement in response consistency when using RAG-based models.

The combination of LangChain’s orchestration capabilities with OpenAI’s GPT-3.5-turbo model enabled the proficient handling of user intent, skilfully routing questions either to the Retrieval-Augmented Generation (RAG) pipeline or to a rule-based fare estimator (Hyperight, 2024). This hybrid system architecture allowed the chatbot to balance flexibility and precision, providing both conversational responses and authoritative results when required.

One of the system’s key strengths lies in its modular design and traceability. Responses were derived from structured input (CSV), which reduced hallucinations and increased transparency, which is an approach recommended by Lewis et al. (2020) for knowledge-grounded AI systems. Additionally, the use of low temperature generation as well as carefully designed prompts enhanced the factual reliability of model outputs, as suggested by Liu (2023)

However, the system had some limitations, particularly in its ability to handle indirect or vague questions posed by users. The absence of conversational memory was a major limitation to its competency to answer follow-up questions in multi-turn conversations. Such issues reflect challenges commonly faced by static data-driven systems and were similarly noted in studies by C4Scale (2024) and Liu (2023).

5.2 Contribution to Knowledge

This research contributes meaningfully to the evolving field of conversational AI by demonstrating that RAG models, traditionally associated with large unstructured datasets (Lewis et al., 2020), can also be readily used for small, structured datasets, such as local transport schedules in CSV format. While most RAG implementations focus on domains like customer service

or document summarization (ChatBees, 2024), this work presents a novel application within the relatively untouched field of static public transportation data.

The project also lays out a hybrid design that combines rule-based fare principles with generative language capabilities. This combination of deterministic tools and AI-driven response generation allows for greater flexibility in user interaction, while maintaining control and transparency in information delivery qualities emphasized by Velaro (2023) as essential for public-facing systems.

In addition, the system architecture, built with LangChain, FAISS, Streamlit, and OpenAI's GPT-3.5-turbo, provides a productive and reproducible model for building cost-effective and scalable transportation chatbots. Its reliance on static data makes it accessible to regions with limited access to real-time APIs or advanced infrastructure, further extending its practical relevance in the context of inclusive and equitable digital services (Sirivastav et al., 2021).

Also, the chatbot in this project makes public transport more accessible to everyone, particularly those who have trouble with regular mobile apps or websites. Studies indicate that elderly individuals commonly have large difficulties with technology, such as not knowing how to use smartphones and not knowing whether they can utilize online services. This makes them have difficulty with receiving important transport information (FANG et al., 2024). Furthermore, AI chatbots have been found to assist elderly individuals in being more self-sufficient and making services less complicated by allowing them to converse naturally rather than going through a difficult app (Santos et al., 2023).

This project enables individuals to utilize transport data. It is inexpensive and requires very few resources, and can be modified to suit various requirements. Individuals can inquire about routes, schedules, and fares without requiring real-time GPS or technical skills. For example, an individual can inquire, "What is the next bus from Hanley to Shelton?" and receive a simple and speedy response based on the data available.

The benefits of this approach are:

- a) Low resource environments are settings that lack sufficient funds or quality facilities to enable live GPS tracking or complex journey planning systems to function.
- b) Foreigners and older users struggle to use normal transport apps as they are complex and use hard words.

- c) Public systems that are replicated for the reason that the design of the chatbot also applies to other modes of transportation (such as trains and ferries) or to other regions such as health and local government inquiries.

The system also reduces the demand for printed timetables, call services and reading timetables manually. It enables transport providers to work effectively and enables more users to use the public services on an equal basis.

5.3 Limitations

Despite its effectiveness, the current model is limited by a number of limitations hindering its usage in more dynamic environments. First, the system operates on static, predetermined data only and cannot accept actual GPS input or real-time modifications to schedules. These limitations curtail its flexibility in circumstances where users require instant notification of delays or changes to routes, a functionality that has been considered critical by Velaro (2023) in government chatbot deployment.

Second, the system struggles with ambiguous and colloquial user queries, particularly when location names do not match the dataset's exact terminology. This limitation reflects similar challenges in structured data RAG models noted by Liu (2023), where semantic retrieval fails without fuzzy matching or NER enhancements.

Third, the lacking session memory makes it difficult to have coherent conversations that last several turns. This is a recognized issue with static RAG systems (Roller et al., 2021). Without conversational memory, users must repeat or rephrase context in each message. Future updates could involve incorporating memory buffers or dynamic memory architectures to improve the flow of conversation (Zhang et al., Adiwardana et al., 2020).

Finally, the chatbot is only built for the English language and is designed for text-based input, thereby restricting its usability for non-English speakers and users with disabilities. Although accessibility features such as voice input and multilingual support are common in commercial bots (Reddy et al., 2023), neither of these features was included in this prototype due to limited factors related to project scope.

5.4 Future Work

To enhance the functionality of the system, several future improvements are recommended. First, integrating real-time transit data such as GPS tracking and live delay notifications would improve the system's relevance for commuters. Accessing such data via APIs from transport authorities could enable dynamic responses and adaptive route guidance, aligning with recommendations by Velaro (2023) and Singhal (2023) for smart mobility platforms.

Second, introducing conversational memory would allow the chatbot to handle multi-turn dialogue, enabling users to ask follow-up questions more naturally. Tools like LangChain's conversation buffer or memory modules can facilitate this, making interactions more intuitive and continuous (ChatBees, 2024).

Third, implementing fuzzy matching or named entity recognition (NER) would help interpret informal queries and better match them with structured data. This solution can address the current system's difficulty in understanding vague stop names or spelling mistakes, a challenge identified by Liu (2023).

Fourth, adding voice interfaces to supplement multilingual support would increase the accessibility of the system, thus expanding its use for more people, such as the visually impaired as well as non-English speakers. According to Reddy et al. (2023) voice voice-enabled interfaces are increasingly essential in the development of a smart city that is inclusive.

Finally, extending the functionality of the chatbot to include other modes of transport, like trains, bikes, and parking options, would make it even more useful as a comprehensive urban trip planning tool. Such integration aligns with the broad visions of smart city plans and would increase its utility for daily urban mobility plans (Singhal, 2023).

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